NATHANIEL McINTIRE

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PROFESSIONAL SUMMARY

Senior aerospace engineering student at NC State University with minors in mathematics, physics, and astrophysics. Experience using MATLAB and ANSYS for engineering analysis, structural testing, and experimental data analysis. Payload and Test/Safety Lead for a lunar rover senior design project and payload team member for NC State's high-powered rocketry program. Brings supervisory experience and a strong interest in structures, materials, and aerodynamics to an entry-level aerospace engineering role.

EDUCATION

NORTH CAROLINA STATE UNIVERSITY **Expected Graduation: May 2027**

Bachelor of Aerospace Engineering | Minors: Astrophysics, Physics, Mathematics

- Relevant Coursework: Aerodynamics I & II, Aerospace Structures I & II, Automatic Controls, Space Flight, Thermodynamics, Numerical Analysis, FEA, Rocket Propulsion, MATLAB

BRUNSWICK COMMUNITY COLLEGE **May 2024**

Associate of Engineering | Associate of Science | Cum Laude Honors

ENGINEERING PROJECTS & DESIGN EXPERIENCE

NASA RASC-AL LUNAR RESOURCE EXPLORATION ROVER - PAYLOAD LEAD **August 2026 - Present**

- Developing a lunar rover concept to map permanently shadowed regions (PSRs) and characterize volatile and mineral resources found in regolith.
- Lead payload development, including the regolith breakup mechanism, selection of resource-analysis instruments, and other sensors needed for operation.
- Coordinate sensor and mechanism requirements with other subteam leads to integrate the payload into the rover design.
- Selected LiDAR for terrain mapping in PSRs, where navigation must operate without sunlight.
- Selected XRF, near-infrared, and mass spectrometry instruments to measure regolith composition and volatile resources.

NC STATE HIGH-POWERED ROCKETRY - IREC PAYLOAD SUBTEAM **August 2026 - Present**

- Support integration and testing of a 4U rover payload designed to deploy autonomously and return to the rocket without GPS for NC State's inaugural International Rocket Engineering Competition (IREC) entry.
- Designing airbrakes to withstand a combined aerodynamic load exceeding 400 lb, approximately 100 lb per brake.
- Using ANSYS to compare airbrake materials under aerodynamic loads at speeds up to Mach 1.2.

COMPOSITE LAMINATE STRESS ANALYSIS **May 2026**

- Evaluated ply-level stresses in composite laminates under multiple loading conditions and transformed stresses between material and global coordinate systems.
- Used MATLAB and classical laminate theory to compare carbon/epoxy stacking sequences, calculate bending stiffness, and predict first-ply failure using the maximum stress criterion.

CONVERGING-DIVERGING NOZZLE DESIGN AND FLOW ANALYSIS **December 2025**

- Designed a converging-diverging nozzle in SolidWorks, manufactured it via 3D printing, and modeled internal pressure and velocity in ANSYS CFD.
- Conducted wind-tunnel testing at NC State and processed experimental data in MATLAB to compare measured and computational flow characteristics.

TECHNICAL SKILLS

Software: MATLAB | Python | C++ | ANSYS | SolidWorks | XFOIL | Excel | KiCad

Engineering Methods: FEA | CFD | Structural Analysis | Composite Stress Analysis | Control System Design | PLC Programming

PROFESSIONAL EXPERIENCE

PRANZO ITALIANO | Server | Raleigh, NC **September 2026 - Present**

BONGIORNO & SON ITALIAN SPECIALTIES | Supervisor | Raleigh, NC **March 2025 - August 2026**

ACE HARDWARE | Supervisor | Raleigh, NC **August 2024 - March 2025**

LOWES FOODS | Guest Services Manager | Wilmington, NC **October 2020 - August 2024**